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Last updated April 2, 2026

Research Interests

(locally) symmetric spaces, spherical and affine buildings, quantum chaos, spectral theory, scattering theory, polytopal geometry, representation theory of real and p -adic groups, automorphic forms, connections with algebraic combinatorics

Academic Positions

Sorbonne University, Institut de Mathématiques de Jussieu-Paris Rive Gauche, Paris, France
Postdoctoral researcher funded by the [National Science Foundation Mathematical Sciences Postdoctoral Research Fellowship](#) and working with [Farrell Brumley](#) in the Automorphic Forms group.
October 2025 – September 2028

Sorbonne University, Institut de Mathématiques de Jussieu-Paris Rive Gauche, CNRS, Paris, France
Postdoctoral researcher funded by the [Cofund MathInGreaterParis Fellowship](#) and working with [Farrell Brumley](#) in the Automorphic Forms group.
October 2024 – September 2025

Paderborn University, Paderborn, Germany
Postdoctoral researcher in the Collaborative Research Center between Bielefeld University and Paderborn University on the thematic program [Integral Structures in Geometry and Representation Theory](#) (Projects [B3](#) and [B4](#)). Member of the [Spectral Analysis group](#) of Tobias Weich.
September 2023 – October 2024

Education

University of Michigan, Ann Arbor, Michigan
Ph.D. in Mathematics
Advised by [Ralf Spatzier](#)
Thesis title: “Quantum Ergodicity on Bruhat-Tits Buildings”
April 2023

Yale University, New Haven, Connecticut
B.A. in Mathematics
May 2017

Budapest Semesters in Mathematics, Budapest, Hungary
Fall 2015 and Spring 2016

Extended Research Visits

Aalto University, Espoo, Finland
Visiting researcher in the group of [Tuomas Sahlsten](#)
Summer 2023

Preprints

4. [Quantum ergodicity on locally symmetric spaces in the Benjamini-Schramm limit](#), with F. Brumley, S. Marshall, and J. Matz. 66 pages. (2026)
3. [Resonances on geometrically finite graphs](#), with C. Arends and T. Weich. 51 pages. (2026).
2. [The discrete wave equation with applications to scattering theory and quantum chaos](#). 46 pages. (2025)
1. [Quantum ergodicity on the Bruhat-Tits building for \$\mathrm{PGL}\(3, F\)\$ in the Benjamini-Schramm limit](#). 71 pages. (2023)

Publications

4. [A degenerate version of Brion's formula](#).
Advances in Mathematics 486 (2026), Paper No. 110732
3. [A geometric perspective on the MSTD question](#), with S. J. Miller.
Discrete and Computational Geometry 62, 832-855 (2019).
2. [The bidirectional ballot polytope](#), with S. J. Miller, C. Sprunger, and R. Van Peski.
Integers 18 (2018), #A81.
1. [Summand minimality and asymptotic convergence of generalized Zeckendorf decompositions](#), with K. Cordwell, M. Hlavacek, C. Huynh, S. J. Miller, and Y. N. T. Vu.
Research in Number Theory (2018) 4: 43.

Awards

NSF Postdoctoral Fellowship (\$190,000)	Fall 2025-Fall 2028
Cofund MathInGreaterParis Fellowship	Fall 2023 - Fall 2025
Arthur Herbert Copeland, Sr. Memorial Scholarship	Summer 2021
Highest Honors in Budapest Semesters in Mathematics	Fall 2015, Spring 2016
MAA Outstanding Presentation Award	August 2015
John Alan Lewis Summer Research Fellowship	Summer 2015

Invited Talks

- [Quantum ergodicity in the Benjamini-Schramm limit for locally symmetric spaces](#)
Analysis & Number Theory Seminar, Tübingen University, April 2026 (planned)
- [Resonances on geometrically finite graphs](#)
Spectra and moduli seminar, Durham University, February 2026
- [Quantum ergodicity in the Benjamini-Schramm limit in higher rank](#)
Math department seminar, Institut Denis Poisson, Université d'Orléans, January 2026
- [The wave equation on trees and buildings](#)
Groups, Geometry, and Dynamics Oberseminar, Karlsruhe Institute of Technology, November 2025
- [Quantum ergodicity in the Benjamini-Schramm limit in higher rank](#)
Analysis seminar, Wrocław University of Science and Technology, October 2025

- *Resonances on geometrically finite graphs*
University of Utah Max Dehn seminar, September 2025
- *Quantum ergodicity in the Benjamini-Schramm limit in higher rank*
University of Utah Math Colloquium, September 2025
- *The multitemporal wave equation on Bruhat-Tits buildings*
University of Utah Representation theory/Number theory seminar, September 2025
- *Quantum ergodicity in the Benjamini-Schramm limit in higher rank*
Aarhus Automorphic Forms Conference, August 2025
- *A degenerate version of Brion's formula*
Combinatorics, arithmetic and geometry seminar, Sorbonne University, March 2025
- *Quantum ergodicity in the Benjamini-Schramm limit in higher rank*
Geometry, groups, and dynamics seminar, ENS de Lyon, February 2025
- *Quantum ergodicity in the Benjamini-Schramm limit in higher rank*
Dynamical systems seminar, Université Sorbonne Paris Nord, February 2025
- *Quantum ergodicity in the Benjamini-Schramm limit in higher rank*
Workshop on automorphic forms and number theory, Sorbonne University, December 2024
- *Quantum ergodicity in the Benjamini-Schramm limit on higher rank real and p -adic locally symmetric spaces*
Geometry & Topology Seminar, Sorbonne University, November 2024
- *Degenerate Brion's formula and applications to symmetric spaces and buildings*
Geometry & Groups Seminar, Heidelberg University, July 2024
- *Quantum ergodicity in the Benjamini-Schramm limit on higher rank real and p -adic locally symmetric spaces*
Dynamics Seminar, Northwestern University, May 2024
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3)$ in the Benjamini-Schramm limit*
Analysis & Number Theory Seminar, Tübingen University, April 2024
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3)$ in the Benjamini-Schramm limit*
New England Dynamics and Number Theory Seminar, April 2024 (virtual talk)
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3)$ in the Benjamini-Schramm limit*
Groups, Dynamics, & Topology Seminar, Jagiellonian University, March 2024
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3)$ in the Benjamini-Schramm limit*
Group Actions Seminar, UC San Diego, February 2024 (virtual talk)
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3, F)$ in the Benjamini-Schramm limit*
Topology/Geometry Seminar, Göttingen University, January 2024
- *Quantum ergodicity on Bruhat-Tits buildings*
Geometric and Functional Analysis Seminar, University of Helsinki, January 2024
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3, F)$ in the Benjamini-Schramm limit*
Number Theory Seminar, Johns Hopkins University, December 2023
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3, F)$ in the Benjamini-Schramm limit*
Number Theory & Representation Theory Seminar, University of Maryland-College Park, December 2023
- *Brion's formula and its applications to analysis on Bruhat-Tits buildings*
Groups & Geometry Seminar, Bielefeld University, December 2023
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3, F)$ in the Benjamini-Schramm limit*
Number Theory Lunch Seminar/Analytic Number Theory & Automorphic Forms Seminar, Max Planck Institute/Bonn University, November 2023
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3, F)$ in the Benjamini-Schramm limit*
Geometric & Harmonic Analysis Seminar, Paderborn University/Aarhus University, October 2023

- *Quantum ergodicity on the Bruhat-Tits building for $PGL(3, F)$ in the Benjamini-Schramm limit*
Buildings 2023, Gießen University, October 2023
- *Quantum ergodicity on Bruhat-Tits buildings*
Mathematical Physics Seminar, Aalto University, August 2023
- *Quantum ergodicity on Bruhat-Tits buildings*
Number Theory & Representation Theory Seminar, University of Wisconsin–Madison, March 2023
- *Quantum ergodicity on Bruhat-Tits buildings of type $\tilde{A}_2$*
RTG Geometry, Dynamics and Topology Seminar, University of Michigan, April 2022

Teaching Experience

Paderborn University

I designed and was the sole instructor for a Master's course titled [Buildings and the Structure of \$p\$ -adic Groups](#) during the summer term of 2024. The class met 3 hours per week for 14 weeks.

University of Michigan

Winter 2023	Calculus II	Primary Instructor
Fall 2022	Calculus II	Primary Instructor
Fall 2021	Calculus III	Lab Instructor (MATLAB)
Fall 2020	Calculus II	Primary Instructor
Fall 2019	Calculus III	Lab Instructor (MATLAB)
Fall 2019	Lie Algebras	Grader
Spring 2019	Ordinary Differential Equations	Lab Instructor (MATLAB)
Winter 2019	Calculus II	Primary Instructor
Fall 2018	Calculus II	Primary Instructor
Winter 2018	Calculus I	Primary Instructor
Fall 2017	Calculus I	Primary Instructor

Outreach & Service

- Animateur at Fête de la Science October 2025
 - Assisted with the animation of a math stand as part of a larger science festival for the general public.
- Admissions Committee for Lab of Geometry at Michigan (LoG(M)) REU Winter 2019 – Fall 2022
 - Lab of Geometry at Michigan (LoG(M)) is a Research Experience for Undergraduates for University of Michigan students which takes place during the Fall and Winter semesters. I served on the admissions committee every term between Winter 2019 and Fall 2022.
- Mentor for Directed Reading Program Fall 2019
 - The Directed Reading Program pairs graduate students with undergraduates, whom they mentor as they work through more advanced mathematical material. I mentored an undergraduate on a project on differential geometry of curves and surfaces.
- Speaker at Michigan Math Circle February 2019
 - Michigan Math Circle is an enrichment program for middle and high school students to get exposed to math outside of the K-12 curriculum. I designed and led two 90 minutes sessions on Zeckendorf decompositions and related topics for high school students.
- Speaker at Michigan Undergraduate Math Club March 2018, April 2019
 - I gave two talks to the University of Michigan undergraduate math club: one on Cohn's irreducibility criterion and the other on integral geometry (Crofton's formula and related topics).

- Co-mentor for LoG(M) REU project Fall 2018
 - I co-mentored three undergraduates on a research project related to translation surfaces along with another graduate student (Matt Stevenson) and a faculty member (Alex Wright, who proposed the project). [Here](#) is the final poster for the project.

Contributed Talks/Posters

- *A degenerate version of Brion's formula*
Winter School and Workshop on Polytopes 2025, Universität Osnabrück, February 2025 (poster)
- *A degenerate version of Brion's formula*
Polytopes and Friends, KTH Stockholm, December 2024 (poster)
- *Degenerate Brion's formula and applications*
60th Sophus Lie Seminar, Paderborn University, September 2024
- *Quantum ergodicity in the Benjamini-Schramm limit in higher rank*
Bridging the Physics and Mathematics of Quantum Many Body Chaos, University of Helsinki, June 2024 (poster)
- *Quantum ergodicity in the level aspect on higher rank real and p -adic locally symmetric spaces*
36th Automorphic Forms Workshop, Oklahoma State University, May 2024
- *Quantum ergodicity on the Bruhat-Tits building for $\mathrm{PGL}(3, F)$ in the Benjamini-Schramm limit*
Summer School: Microlocal and Probabilistic Methods in Geometry and Dynamics, Jussieu, France, July 2023 (poster)
- *Quantum ergodicity on Bruhat-Tits buildings of type $\tilde{A}_2$*
2022 Midwest Representation Theory Conference, University of Michigan, March 2022
- *On summand minimality of generalized Zeckendorf decompositions*
Joint Math Meetings, Atlanta, GA, January 2017
- *A geometric perspective on the MST question*
INTEGERS 2016, University of West Georgia, October 2016
- *Generalized numerical semigroups of minimal embedding dimension*
MAA MathFest 2015, Washington, D.C., August 2015
- *Some results on two-lifts of graphs*
Joint Math Meetings 2015, San Antonio, TX, January 2015

Learning Seminar Talks

- *Goldman-Iwahori spaces*
Groupe de Travail on Bruhat-Tits buildings and Berkovich spaces, IMJ-PRG, February 2026
- *The Morgan-Shalen compactification of character varieties: valuations and Λ -trees*
Paderborn/Leipzig learning seminar on character varieties (virtual), December 2025 (two part talk)
- *The polynomial method for strong convergence*
Groupe de Travail on strong convergence, Sorbonne University, November and December 2025 (two part talk)
- *Introduction to Fredholm operators*
Summer school on loops groups and Kac-Moody groups, Willebadessen, September 2025
- *Non-uniform tree lattices*
Learning Seminar on Lattices in Lie Groups, Paderborn University, July 2024
- *Anosov representations on Bruhat-Tits buildings*
Patterson-Sullivan Theory/Anosov Representations Learning Workshop, Paderborn University, March 2024
- *The local Weyl law*
Microlocal Analysis Learning Seminar, Aalto University/University of Helsinki, November 2023

- *L^2 -theory and ellipticity*
Microlocal Analysis Learning Seminar, Aalto University/University of Helsinki, October 2023
- *Analysis on Bruhat-Tits buildings*
Geometry Seminar, University of Michigan, February 2023
- *Geometry of Bruhat-Tits buildings*
RTG Geometry, Dynamics and Topology Seminar, University of Michigan, February 2023
- *Quantum ergodicity in the Benjamini-Schramm limit*
RTG Geometry, Dynamics and Topology Seminar, University of Michigan, February 2023
- *The Harish-Chandra isomorphism and the Satake isomorphism for $SL(2)$*
RTG Representation Theory Seminar, University of Michigan, March 2022
- *Applications of representation theory of $SL(2, \mathbb{R})$ to dynamics*
RTG Representation Theory Seminar, University of Michigan, December 2021
- *The Fell topology*
Student Dynamics/Geometry/Topology Seminar, University of Michigan, March 2021
- *Quantum ergodicity on graphs*
Student Analysis Seminar, University of Michigan, March 2021
- *Some analogies between hyperbolic surfaces and regular graphs*
Student Dynamics/Geometry/Topology Seminar, University of Michigan, November 2020
- *What is quantum ergodicity?*
Student Analysis Seminar, University of Michigan, October 2020
- *General relativity for mathematicians*
Student Geometry/Topology Seminar, University of Michigan, November 2019
- *Coxeter groups and buildings*
Student Geometry/Topology Seminar, University of Michigan, February 2019
- *Spectral graph theory*
Student Combinatorics Seminar, University of Michigan, September 2018
- *Introduction to Ehrhart theory*
Student Combinatorics Seminar, University of Michigan, March 2018

Outreach Talks

- *Bruhat-Tits buildings*
Mathjeunes Seminar, Sorbonne University, November 2024
- *The Poincare series of a Coxeter group and its applications*
Paderborn/Bielefeld CRC Graduate Seminar, Paderborn University, May 2024
- *Quantum ergodicity on manifolds and graphs*
Physics Graduate Student Symposium, University of Michigan, July 2021
- *Crofton's formula, Buffon's needle, and the isoperimetric inequality*
Michigan Undergraduate Math Club, University of Michigan, April 2019
- *Exotic number systems*
Michigan Math Circle, February 2019 (two part talk)
- *Cohn's irreducibility criterion*
Michigan Undergraduate Math Club, University of Michigan, March 2018

Other Conferences/Workshops Attended

- Tropical Geometry, Moduli Spaces and Locally Symmetric Varieties
Bielefeld University, March 2026
- Lie theory, Spectra, and Dynamics
Paderborn University, September 2025
- Integral Structures in Geometry and Representation Theory
Paderborn University, September 2024
- Building Bridges: 6th EU/US Summer School & Workshop on Automorphic Forms and Related Topics
CIRM, September 2024
- Automorphic Forms in Budapest 2024
Rényi Institute, August 2024
- Zeta Functions, Dynamics, and Analytic Number Theory
Göttingen University, March 2024
- New Perspectives in the Analytic Theory of Automorphic Forms
Clay Math Institute, University of Oxford, September 2023
- Durham Symposium 2023: Spectral Gaps
Durham University, August 2023
- Summer School on High-Dimensional Expanders
Ghent University, May 2023
- Dynamics, Rigidity and Arithmetic in Hyperbolic Geometry
ICERM, Brown University, May 2023
- Laplacians on Random Hyperbolic Surfaces and on Random Graphs
Northwestern University, May 2022
- Spectra and Dynamics on (Locally) Symmetric Spaces
Universität Paderborn, February 2022
- Microlocal Analysis: Theory and Applications
University of Montreal, Summer 2021 (MSRI virtual summer school)
- Dynamics and Geometry Online Summer School
Heilbronn Institute, Bristol University, June 2021 (virtual summer school)
- Midwest Dynamical Systems Conference
University of Illinois at Chicago, November 2019
- Regional Workshop in Quantitative Geometry & Topology
The Ohio State University, April 2019
- Graduate Student Topology and Geometry Conference
University of Illinois at Urbana-Champaign, April 2019
- Park City Math Institute Undergraduate Summer School on Random Matrix Theory
Park City, Utah, July 2017